



**COMPETENCY STANDARD
FOR
CNC MACHINE OPERATION
(*LIGHT ENGINEERING SECTOR*)**

**Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance**

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The Competency Standards for CNC Machine Operation is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment and subsequently, they would be qualified and settled for a relevant job.

The document was developed under the Skills for Employment Investment Program (SEIP) and subsequently reviewed and updated with the engagement of occupation-specific industry experts to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, training and assessment may be conducted at the workplace, at training institute or during regular work, or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at Glance:

Generic (Basic) Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-CNC-01-G	Perform Computations Using Basic Mathematical Concepts	1. Identify calculation requirements in the workplace 2. Identify and Select appropriate mathematical methods. 3. Use tool/instrument to perform calculations	10
SICIP-LE-CNC-02-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-LE-CNC-03-G	Communicate in English in the Workplace	1. Read and understand workplace documents in English 2. Prepare simple workplace documents in English 3. Listen and comprehend to English conversations 4. Perform conversations in English language	5
SICIP-LE-CNC-04-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member. 4. Solve problems as a team member	5
Total Hour			30

Sector Specific (Common) Competencies (40 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-CNC-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	04
SICIP-LE-CNC-02-S	Translate Manuals, Mechanical Drawings, Specifications and the way of Estimation	1. Understand symbols, mechanical drawing, specifications and way of estimation 2. Interpret manuals, mechanical drawings specific to light engineering works 3. Store manuals, drawings and plans	12
SICIP-LE-CNC-03-S	Work with Hand Tools, Power Tools	1. Inspect hand tools and power tools for usability 2. Use hand tools properly and safely 3. Operate power tools properly and safely 4. Clean/maintain hand tools and power tools after use	12
SICIP-LE- CNC-04-S	Carry Out Measurements and Calculations	1. Check usability of measuring devices 2. Carry out accurate work measurements 3. Execute simple work calculations 4. Clean and maintain measuring instruments	12
Total Hour			40

OCCUPATION SPECIFIC (CORE) COMPETENCIES (290 HRS.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-CNC-01-O	Understand mechanical drawing	1. Interpret mechanical drawing 2. Create mechanical drawing	40
SICIP-LE-CNC-02-O	Carry out CNC lathe machine operations	1. Prepare CNC lathe machine 2. Edit NC-Code of CNC lathe Machine 3. Cut sample work piece 4. Perform CNC lathe machine operations 5. Check and measure work piece 6. Maintain tools, equipment, machinery and systems	90
SICIP-LE-CNC-03-O	Carry out CNC milling machine operations	1. Prepare CNC milling machine 2. Edit NC-Code of Milling Machine 3. Perform CNC milling machine operations 4. Check and measure work piece 5. Maintain tools, machinery and systems	120
SICIP-LE-CNC-04-O	Carry out CNC wire cut machine operations	1. Set- up machine, wire and work piece 2. Load program to wire cut machine 3. Perform CNC wire cut operations 4. Clean and maintain wire cut machine	40
Total Hours			290

COMPETENCY STANDARD: CNC MACHINE OPERATION

A. The Generic (Basic) Competencies

Unit of Competency: PERFORM COMPUTATIONS USING BASIC MATHEMATICAL CONCEPTS	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-CNC-01-G
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform computations using basic mathematical concepts in workplace. It specifically includes the tasks of identifying calculation requirements in the workplace, identifying and selecting appropriate mathematical methods, and using appropriate tools and instruments to perform calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify calculation requirements in the workplace	1.1 <u>Calculation requirements</u> are identified 1.2 <u>Workplace information</u> are interpreted
2. Identify and select appropriate mathematical methods	2.1 <u>Appropriate calculation method</u> is identified 2.2 Appropriate calculation method is selected
3. Use tool/instrument and perform calculations	3.1 Appropriate <u>tools and instruments</u> are identified 3.2 Computation is made

Range of variables:

Variable	Range
	May include but not limited to:
1. Calculation requirements.	1.1 Area 1.2 Volume 1.3 Measurements: Length/Width/Height/thickness/diameter 1.4 Weight and density Calculation
2. Workplace information	2.1 Drawing 2.2 Part drawing 2.3 Verbal instructions 2.4 Job order and Job sheet
3. Appropriate calculation method	3.1 Addition, Subtraction, Division and Multiplication 3.2 Conversion of measurement 3.3 Percentage and ratio calculation 3.4 Geometric Formulae (Square, Rectangle, Circle, Sphere etc)
4. Tools/instruments	4.1 Calculator 4.2 Computer

Curricular Content Guide

1. Underpinning Knowledge	1.1 Numerical concept, identification of calculation requirements 1.2 Basic mathematical methods such as addition, subtraction, multiplication, division, percentage, conversion, ratio and basic geometry formulas. 1.3 Mathematical language, symbols.
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	1.4 Measuring units
2. Underpinning Skills	2.1 Computing basic mathematics. 2.2 Practicing conversion of measurements 2.3 Calculating areas, volume etc using formulae
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect to peers, sub-ordinates and seniors in workplace 3.5 Environmental concern 3.6 Sincerity and honesty
4. Resource Implications	The following resources must be provided. 4.1 Stationeries 4.2 Consumables 4.3 Calculators 4.4 Computers 4.5 Measuring tape

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified calculation requirements from workplace information. 1.2 Selected appropriate method to carry out the calculation requirements. 1.3 Completed calculations using appropriate tools/instruments.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Oral questioning 2.3 Demonstration.
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-CNC-02-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards. 3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.

4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace 3.3 Promptness in carrying out activities 3.4 Tidiness and timeliness 3.5 Respect of peers, sub-ordinates and seniors in workplace 3.6 Environmental concern 3.7 Sincere and honest to duties
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Practiced personal health and safety procedures 1.4 Reported incidents arising from hazards and risks to authority. 1.5 Implemented plans and procedures to respond emergency 1.6 Applied basic first aid procedure.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test

	2.2 Demonstration 2.3 Oral questioning 2.4 Interview
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: COMMUNICATE IN ENGLISH IN THE WORKPLACE	Nominal Duration: 05 hrs.	Unit Code: SICIP-LE-CNC-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to communicate in English in the workplace. It specifically includes the tasks of reading and understanding workplace documents in English, preparing simple workplace documents in English, listening and comprehending to English conversations and performing conversations in English.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Read and understand workplace documents in English	1.1 Workplace documents are read and understood 1.2 Visual information is interpreted.
2. Prepare simple workplace documents in English	2.1 Simple <u>routine workplace documents</u> are prepared using key words, phrases, simple sentences 2.2 <u>Visual aids</u> are prepared and submitted. 2.3 Key information is written in the appropriate places in standard forms.
3. Listen and comprehend to English conversations	3.1 Active listening is demonstrated. 3.2 English conversation is comprehended and shared
4. Perform conversations in English language	4.1 Conversation is performed in English with peers, customers and management as per the workplace standard.

Range of Variables

Variable	Range
	May Include but not limited to:
1. Routine workplace documents	1.1 Agenda 1.2 CV/Resume 1.3 Simple reports such as progress and incident reports 1.4 Job sheets 1.5 Operational manuals 1.6 Brochures and promotional material 1.7 Visual and graphic materials 1.8 Standards 1.9 OSH information 1.10 Signs
2. Visual aids	2.1 Maps 2.2 Diagrams 2.3 Forms 2.4 Labels 2.5 Graphs 2.6 Charts

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Interpret workplace documents in English 1.2 Write simple routine workplace documents in English 1.3 Listen to conversation in English 1.4 Perform conversation in English 1.5 Interaction skills (i.e., teamwork, interpersonal skills, etc.)
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	1.6 Job roles, responsibilities and compliances
2. Underpinning Skills	2.1 Ability to read and understand workplace documents in English by using appropriate vocabulary and grammar, standard spelling and punctuation 2.2 Ability to write simple routine workplace documents in English such as: Schedules and agenda, job sheets, operational manuals and brochures and promotional material 2.3 Ability of listening in English and interpreting 2.4 Ability to perform conversation in English with peers, customers and management 2.5 Work effectively with others 2.5.1 Listening and questioning skills 2.5.2 Ability to follow simple directions
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect of peers, sub-ordinates and seniors in workplace 3.5 Environmental concern 3.6 Sincere and honest to duties
4. Resource Implications	The following resources must be provided: 4.1 Work place Procedure 4.2 Materials relevant to the proposed activity 4.3 All tools, equipment, material and documentation required. 4.4 Relevant specifications or work instructions

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Conversed in English with peers and customers. 1.2 Made reports of workplace documents in English.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Demonstration 2.3 Oral questioning 2.4 Interview
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: OPERATE IN A SELF-DIRECTED TEAM	Nominal Duration: 05 hrs.	Unit Code: SICIP-LE-CNC-04-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members.	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member.	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practice. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range
	May Include but not limited to:
1. Forms of communication	1.1 Agenda 1.2 CV 1.3 Simple reports such as progress and incident reports 1.4 Job sheets 1.5 Operational manuals 1.6 Brochures and promotional material 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning. 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Support mechanism to other members of team to ensure achievements of goals. 1.10 Methods of identifying current and potential problems faced by a team 1.11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice. 2.8 Team member's support to other members is appreciated and valued 2.9 Identifying solutions to the problem 2.10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Teamwork 3.2 Promptness in carrying out activities. 3.3 Tidiness and timeliness. 3.4 Respect of peers, sub-ordinates and seniors in workplace. 3.5 Sincere and honest to duties
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Pens 4.3 Papers 4.4 Work books 4.5 Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified team goals and work processes 1.2 Communicated and cooperated with team members. 1.3 Worked as a team member 1.4 Solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to:

	2.1 Written test 2.2 Demonstration 2.3 Oral questioning
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

B. The Sector Specific (Common) Competencies

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 04 hrs.	Unit Code: SICIP-LE-CNC-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during light engineering activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in light engineering sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 <u>Recyclable and non-recyclable items</u> are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste</u> are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 use of sustainable materials with low environmental impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution

	2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing Energy-Efficient Equipment 3.2 Adopting Renewable Energy Sources 3.3 Implementing Site Protection Measures 3.4 Using Reusable Materials 3.5 Choosing Sustainable Materials 3.6 Using Noise-Reducing Equipment and scheduling work appropriately 3.7 Optimizing Logistics and Delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Light Engineering sector 1.2 Key terms and symbols related to environmental sustainability in Light Engineering drawings 1.3 Sources of environmental impacts in Light Engineering 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in Light Engineering activities 2.2 Applying methods to minimize resource use in the workplace 2.3 Implementing waste management practices 2.4 Following procedures to safely dispose of hazardous materials 2.5 Documenting water, energy, and material consumption in the workplace 2.6 Performing green practices in personal and professional activities

3. Underpinning Attitudes	3.1 Commitment to sustainable and eco-friendly Light Engineering practices 3.2 Active participation in reducing resource use and waste generation 3.3 Eagerness to adopt green habits in professional life 3.4 Promptness in implementing sustainable practices in Light Engineering activities 3.5 Tidiness and concern for environmental cleanliness 3.6 Compliance with safety standards and environmental regulations
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waster segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in Light Engineering 1.2 Identified sources of environmental impacts 1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: TRANSLATE MANUALS, MECHANICAL DRAWINGS, SPECIFICATIONS AND ESTIMATE	Nominal Duration: 12 hrs.	Unit Code: SICIP-LE-CNC-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to translate manuals, schematic drawings, specifications and work estimation. It specifically includes the tasks of understanding symbols, mechanical drawing, specifications, and work estimation, interpreting manuals, drawings specific to Engineering works and storing manuals, drawings and plans.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand symbols, mechanical drawing, specifications and the way of estimation	1.1 Symbols are identified and interpreted 1.2 Drawing and specifications are identified 1.3 Way of estimation according to drawing and specifications is demonstrated
2. Interpret manuals, mechanical drawings specific to engineering works	2.1 <u>Manuals</u> and <u>mechanical drawings</u> are correctly recognized from manuals, designs and plans. 2.2 Terms and abbreviations in drawings are recognized. 2.3 <u>Signs and symbols</u> in the drawing are interpreted.
3. Store manuals, drawings and plans	3.1 Manuals, drawings, and plans are collected and packed. 3.2 Manuals, drawings and plans are stored to prevent damage, and ready access and updating of information when required.

Range of Variables

Variable	Range
	May include but not limited to:
1. Manuals	1.1 Manufacturer's Specification Manual 1.2 Repair Manual 1.3 Maintenance Procedure Manual 1.4 Periodic Maintenance Manual 1.5 Quality Manual 1.6 Instruction Manual
2. Mechanical Drawings	2.1 Drawings 2.2 Sketches
3. Signs and symbols	3.1 All signs and symbols associated with the mechanical components and devices

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types of Manuals 1.2 Identification of Signs and Symbols 1.3 Interpretation of signs and symbols
2. Underpinning Skills	2.1 Checking version and date of the manual 2.2 Identifying relevant drawings and specifications correctly 2.3 Identifying terms and abbreviations 2.4 Identifying signs and symbols 2.5 Interpreting drawings and specifications 2.6 Storing manuals
3. Underpinning Attitudes	3.1 Eagerness to learn

	3.2 Orderliness 3.3 Resourcefulness
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering manuals and literatures 4.3 Pens 4.4 Papers 4.5 Work books

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Checked version and date of the manual 1.2 Identified relevant drawings and specifications correctly. 1.3 Identified terms and abbreviations. 1.4 identified Signs and symbols. 1.5 Interpreted drawings and specifications.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: WORK WITH HAND TOOLS AND POWER TOOLS	Nominal Duration: 12 hrs.	Unit Code: SICIP-LE-CNC-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work with hand tools and power tools properly. It specifically includes the tasks of inspecting hand tools and power tools for usability, using hand tools properly and safely, operating power tools properly and safely and cleaning/maintaining hand tools and power tools after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect hand tools and power tools for usability	1.1 Appropriate tools are selected. 1.2 Application of tools to job requirement is determined. 1.3 Usability of tools are checked and verified. 1.4 <u>Hand tools</u> and <u>power tools</u> are prepared. 1.5 Sources of power supply for power tools identified.
2. Use hand tools properly and safely	2.1 Appropriate hand tool for the job is used. 2.2 Proper and safe use/operation is applied in the different types of hand tools. 2.3 <u>Safety precautions</u> is observed when using hand tools. 2.4 Unsafe or faulty tools are identified and marked for repairing
3. Operate power tools properly and safely	3.1 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.2 Proper sequence of operation is applied in using power tools to produce results. 3.3 Power tools are used safely in accordance to manufacturer's operating specification.
4. Clean/maintain hand tools and power tools after use	4.1 Dust and foreign materials are removed from power tools in accordance with workplace standard. 4.2 Scraps or wastages are removed and cleaned 4.3 Condition of tools is checked after use. 4.4 Appropriate lubricant is applied after use and prior to storage. 4.5 <u>Measuring tools</u> are checked and calibrated. 4.6 Defective tools, instruments, power tools and accessories are inspected and corrected or replaced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Hand tools	1.1 Screwdrivers 1.2 Pliers 1.3 Wire Strippers 1.4 Fish Wire (for pulling wire through conduit) 1.5 Leveler 1.6 Tester 1.7 Allen key 1.8 Nut Drivers 1.9 Crimping Tool (for wire terminals) 1.10 Flashlight or Headlamp 1.11 Hacksaw (for cutting conduit)

	1.12 Chisels and Hammer 1.13 Adjustable wrench 1.14 File 1.15 Socket wrench 1.16 Ratchet
2. Power tools	2.1 Power drills 2.2 Nail guns 2.3 Angle grinders and Grinders 2.4 Pneumatic wrenches 2.5 Pedestal drills
3. Safety precautions	3.1 Use of appropriate PPEs 3.2 Proper hand, feet and eye coordination 3.3 Working environment 3.4 Safe operating condition of hand tools and power tools 3.5 Awareness to OHS requirements
4. Measuring instruments	4.1 Tachometer 4.2 Vernier Calipers 4.3 Calipers 4.4 Micrometer 4.5 Bevel Protractor

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of tools, functions and use 1.2 Types of Hand tools and their proper use and techniques 1.3 Types of Power tools, use and safe handling method 1.4 Technical application of tools 1.5 Procedures in the use of hand tools and power tools 1.6 Policies and procedures for occupational health and safety 1.7 Use of PPE 1.8 Handling of tools and equipment 1.9 Reporting and documentation 1.10 Preventive maintenance 1.11 Methods and techniques 1.12 Quality procedures 1.13 Waste management 1.14 Storage procedures
2. Underpinning Skills	2.1 Using appropriate hand tool for the job 2.2 Observing safety precautions when using hand tools 2.3 Using power tools correctly and safely in accordance to manufacturer's operating specification. 2.4 Checking condition of tools after use 2.5 Applying appropriate lubricant on hand tools and power tools after use and prior to storage 2.6 Inspecting and correcting or replacing defective tools, instruments, power tools and accessories 2.7 Storing Tools and power tools safely in appropriate location
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Environmental concerns 3.3 Eagerness to learn 3.4 Tidiness and timeliness 3.5 Concern to proper use of tools 3.6 Orderliness
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools

	4.3 Pens 4.4 Papers 4.5 Work books 4.6 Operation and maintenance manuals
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Used appropriate hand tools for the job. 1.2 Observed safety precautions when using hand tools. 1.3 Used power tools safely in accordance to manufacturer's operating specification. 1.4 Cleaned and maintained hand tools and power tools after use and prior to storage. 1.5 Inspected and corrected or replaced defective tools, instruments, power tools and accessories. 1.6 Stored tools and power tools safely in appropriate location.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: CARRY OUT MEASUREMENTS AND CALCULATIONS	Nominal Duration: 12 hrs.	Unit Code: SICIP-LE-CNC-04-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry-out measurements and calculations. It specifically includes the tasks of checking usability of measuring devices, carrying out accurate work measurements, executing simple work calculations and cleaning and maintaining measuring instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check usability of measuring devices	1.1 Appropriate <u>measuring device</u> is selected for the job. 1.2 Applications of measuring device is determined. 1.3 Usability of measuring device is checked and verified. 1.4 Measuring device is prepared.
2. Carry out accurate work measurements	2.1 Measurements are obtained using appropriate measuring device. 2.2 <u>Systems of measurements</u> are identified and converted where necessary. 2.3 Results are confirmed and recorded.
3. Execute simple work calculations	3.1 Simple calculations involving <u>four basic mathematical calculations</u> are executed. 3.2 Other operations are used to complete tasks in Light Engineering works 3.3 Appropriate formulas for calculating quantities of materials are selected. 3.4 Calculations are performed and verified. 3.5 Material quantities are calculated if required. 3.6 Results are interpreted and communicated to authority.
4. Clean and maintain measuring instruments	4.1 Dust and foreign materials are removed from measuring instrument 4.2 Condition of instrument is checked 4.3 Appropriate lubricant is applied after use and prior to storage 4.4 Measuring instruments are checked and calibrated. 4.5 Instrument is stored in accordance to workplace procedure

Range of Variables

Variable	Range
	May include but not limited to:
1. Measuring device	1.1 Slide calipers 1.2 Steel tape measure 1.3 Steel rule 1.4 Square 1.5 Tri-square 1.6 Feeler gauges 1.7 Water level 1.8 Hose level 1.9 Thermometers 1.10 Protractor
2. Systems of measurements	2.1 Metric system 2.2 FPS system
3. Four basic mathematical	3.1 Area Calculation

Calculation	3.2 Volume Calculation 3.3 Weight Calculation 3.4 Angle Calculation
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Types and principles of operation of measuring devices 1.2 The ISO standard of measurements 1.3 Methods of measurement and calculation 1.4 Fraction and decimals 1.5 Linear measurement 1.6 Units of conversion and conversion factors in measurements 1.7 Dimensioning and fits and tolerances 1.8 Calculating ratio and proportion 1.9 Care in the use of measuring devices
2. Underpinning Skills	2.1 Selecting appropriate measuring device for the job 2.2 Checking and verifying usability of measuring device 2.3 Obtaining measurements using appropriate measuring device. 2.4 Confirming measurements and recording results 2.5 Carrying out simple calculations involving four basic mathematical operations 2.6 Calculating material quantities 2.7 Interpreting and communicating results to authority 2.8 Cleaning and storing measuring instruments
3. Underpinning Attitudes	3.1 Cleanliness/tidiness 3.2 Commitment to occupational health and safety practices 3.3 Environmental concerns 3.4 Eagerness to learn 3.5 Timeliness and orderliness 3.6 Respect for rights of peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of measuring and checking tools/instruments 4.3 Pens 4.4 Papers 4.5 Work books 4.6 Measuring tools' operating and maintenance manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Selected appropriate measuring device for the job. 1.2 Checked and verified usability of measuring device. 1.3 Obtained measurements using appropriate measuring device. 1.4 Confirmed measurements and recorded results. 1.5 Carried out Simple calculations involving four basic mathematical operations. 1.6 Calculated material quantities. 1.7 Interpreted and communicated Results to authority.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training

	module.
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C. Occupation-specific Competencies

Unit of Competency: UNDERSTAND MECHANICAL DRAWING	Nominal Duration: 40 hrs.	Unit Code: SICIP-LE-CNC-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret mechanical drawing, create drawing. It specifically includes interpreting and creating mechanical drawing.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret mechanical drawing	1.1 Different types of <u>mechanical drawings</u> are interpreted. 1.2 Different <u>views of mechanical drawing</u> are identified and described. 1.3 Necessity of putting dimensions in different drawings are comprehended.
2. Create mechanical drawing	2.1 Drawing is created accurately. 2.2 Components, objects, parts, materials and assemblies are identified, drawn and described. 2.3 Base line or datum points are specified, as required. 2.4 Instructions are included in drawing. 2.5 <u>Tolerance</u> , limits and fits are identified in technical drawing.

Range of Variables

Variable	Range
	May include but not limited to:
1. Mechanical drawings	1.1 2D 1.1.1 Line 1.1.2 Rectangle 1.1.3 Circle 1.1.4 Ellipse 1.1.4.1 Polygon 1.1.5 Arc 1.2 3D 1.2.1 Block 1.2.2 Sphere 1.2.3 Pyramid 1.2.4 Taurus
2. Views of mechanical drawing	Drawing technique includes: 2.1 Top and Bottom View 2.2 Right and Left View 2.3 Front and View 2.4 Isometric View 2.5 Oblique View 2.6 Hidden view 2.7 Perspective view
3. Tolerance	3.1 General 3.2 Angular 3.3 Geometric

Circular Content Guide

1. Underpinning knowledge	1.1 Drawing tools and supplies 1.2 Drawing symbols 1.3 Drawing different Views 1.4 Dimensioning techniques 1.5 Tolerance, limits and fits 1.6 Engineering materials
2. Underpinning skills	2.1 Interpreting mechanical drawing 2.2 Creating mechanical drawing
3. Underpinning attitudes	3.1 Tidy and punctual 3.2 Prompt in carrying out activities 3.3 Sincere and honest concerning duties 3.4 Active on teamwork 3.5 Eager to learn 3.6 Concerned for proper use of tools 3.7 Concerned about the work environment 3.8 Committed to occupational health and safety practices
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Drawing tools and supplies 4.3 Specimen parts and components 4.4 Drawings and sketches 4.5 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Interpreted mechanical drawing 1.2 Created mechanical drawing
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Oral test 2.3 Observation 2.4 Demonstration
3. Context of assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: CARRY OUT CNC LATHE MACHINE OPERATIONS	Nominal Duration: 90 hrs.	Unit Code: SICIP-LE-CNC-02-O
Unit Descriptor: This unit of competency covers the skills, knowledge and attitudes required to carry out CNC lathe machine operations. It specifically includes preparing CNC lathe machine, editing NC-Code of CNC lathe Machine, cutting sample work piece, performing CNC lathe machine operations, checking and measuring work piece, maintaining tools, equipment, machinery and systems.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare CNC lathe machine	1.1 Lubricant, air & hydraulic pressure and coolant are checked as per manufacturer's specification. 1.2 <u>Cutting tools</u> are set according to required sequence of operation. 1.3 <u>Clamping devices</u> are set and tightened according to standard operating procedures. 1.4 <u>Tool set-up</u> is performed as per standard operating procedures. 1.5 Work piece is mounted and centered on clamping device to required level of accuracy as per workplace procedures 1.6 Main switch and emergency switch are turned on and voltage stabilizer are checked 1.7 Machine zero point is checked 1.8 Program is downloaded and transferred the program into the machine using appropriate device.
2. Edit NC-Code of CNC lathe Machine	2.1 Functions of commonly used G-Code commands are explained for lathe operations 2.2 Functions of commonly used M-Code commands are explained for lathe operations 2.3 Errors in G-Code and M-Code are identified and adjusted
3. Cut sample work piece	3.1 Dry run is performed in accordance with the desired tool movement. 3.2 Work piece is cut as programmed. 3.3 Work piece is checked and measured using appropriate <u>measuring tools</u> . 3.4 Program is adjusted and tool parameters are corrected as required.
4. Perform CNC lathe machine operations	4.1 Work piece is mounted as per standard operating procedures. 4.2 <u>CNC lathe operations</u> are carried out to produce component as per program. 4.3 <u>Corrective measures</u> are performed, if necessary.
5. Check and measure work piece	5.1 Work piece is checked and measured against specification using appropriate methods and measuring tools. 5.2 Defective work pieces are marked, recorded and reported for proper action.
6. Maintain tools,	6.1 Proper shutdown is carried out in accordance with standard

equipment, machinery and systems	operating procedure. 6.2 Ensure security of data, including regular back-ups and virus checks are performed as per standard operating procedure. 6.3 Basic file maintenance procedures are carried out in line with the standard operating procedure. 6.4 Systems and workplace are cleaned according to worksite procedures. 6.5 CNC lathe machine is cleaned and maintained as per standard operating procedure. 6.6 Tools, equipment, machinery and <u>materials</u> are cleaned and stored safely.
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Range of Variables

Variable	Range
1. Cutting tools	May include but not limited to: 1.1 Facing 1.2 Turning 1.3 Grooving 1.4 Drilling 1.5 Threading 1.6 Parting-off 1.7 Taping 1.8 Boring 1.9 Knurling 1.10 Honing 1.11 Finishing
2. Clamping devices	2.1 Three jaw chuck 2.2 Four jaw chuck 2.3 Collet chuck 2.4 Live center 2.5 Bar feeder 2.6 Part catcher 2.7 Tool center
3. Tool set-up	3.1 Scratch method 3.2 Tool-setting device method
4. Measuring tools	4.1 Vernier caliper (digital or read out) 4.2 Micrometer (digital or read out) 4.3 Gages (thread, drill, surface comparator / roughness tester, radius, screw pitch, taper)
5. CNC lathe operations	5.1 Facing 5.2 Turning 5.3 Tapper Turning 5.4 Threading 5.5 Drilling 5.6 Boring 5.7 Taping 5.8 Parting-off
6. Corrective measures	6.1 Cutting tool changes 6.2 Adjustment of tool offset 6.3 Adjustment of cutting speed and feed rate

7. Materials	7.1 Aluminum 7.2 Brass 7.3 Copper 7.4 Steel 7.5 Titanium 7.6 Nickel 7.7 Thermoset plastics 7.8 Nylon
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Circular Content Guide

1. Underpinning knowledge	1.1 Job requirements and work piece specifications 1.2 Types of cutting tools and materials 1.3 Procedure for checking lubricant and coolant 1.4 Procedure for checking air and hydraulic pressure 1.5 NC-Code commands 1.6 Procedure for running CNC lathe machine 1.7 Uses of measuring tools for corrective measures
2. Underpinning skills	2.1 Establishing job requirements and work piece specifications 2.2 Identifying and selecting appropriate cutting tools and attachments 2.3 Editing NC-Code commands 2.4 Performing dry run 2.5 Cutting work piece 2.6 Setting-up work to using appropriate tools and clamping device 2.7 Setting work to avoid distortion on release of clamping devices 2.8 Selecting, calculating and using appropriate speed, feed and depth of cut 2.9 Planning and sequencing of work 2.10 Checking conformance to specifications 2.11 Performing CNC lathe machine operations 2.12 Checking and measuring work piece against specifications within specified tolerances 2.13 Performing any necessary corrective measures
3. Underpinning attitudes	3.1 Prompt in carrying out activities 3.2 Tidy and punctual 3.3 Eager to learn 3.4 Active on teamwork 3.5 Sincere and honest concerning duties 3.6 Concerned for proper use of tools 3.7 Concerned about the work environment 3.8 Committed to occupational health and safety practices
4. Resource implications	The following resources must be provided: 4.1 Workplace 4.2 Computer/laptop/notebook 4.3 Software 4.4 Personal protective equipment (PPE) 4.5 CNC lathe tools and attachments 4.6 CNC lathe machine 4.7 Drawings, specifications and work instructions 4.8 Materials

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Prepared CNC lathe machine 1.2 Cut sample work piece 1.3 Set-up workpiece using appropriate tools and clamping devices 1.4 Performed CNC lathe machine operations
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Oral test 2.3 Observation 2.4 Demonstration
3. Context of assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: CARRY OUT CNC MILLING MACHINE OPERATIONS	Nominal Duration: 120 hrs.	Unit Code: SICIP-LE-CNC-03-O
Unit Descriptor: This unit of competency covers the skills, knowledge and attitudes required to carry out CNC milling machine operations. It specifically includes preparing CNC milling machine, editing NC-Code of Milling Machine, performing CNC milling machine operations, checking and measuring work piece, maintaining tools, machinery and systems.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare CNC milling machine	1.1 Lubricant, air & hydraulic pressure and coolant are checked as per manufacturer's specification. 1.2 <u>Cutting tools and instruments</u> are set according to required sequence of operation. 1.3 <u>Clamping devices</u> are set and tightened according to standard operating procedures. 1.4 Tool setting method is applied as per standard operating procedures. 1.5 Work piece is mounted and centered on clamping device to required level of accuracy as per workplace procedures 1.6 Main switch and emergency switch are turned on 1.7 Machine zero point is checked 1.8 Program is downloaded and transferred the program into the machine using appropriate device.
2. Edit NC-Code of Milling Machine	2.1 Functions of commonly used G-Code commands are explained for milling operations 2.2 Functions of commonly used M-Code commands are explained for milling operations 2.3 Errors in G-Code and M-Code are identified and adjusted
3. Perform CNC milling machine operations	3.1 Work piece is mounted as per standard operating procedures. 3.2 Different <u>stages of milling</u> for a job are carried out 3.3 <u>CNC milling operations</u> are carried out to produce component as per program. 3.4 Corrective measures such as tool grinding are performed.
4. Check and measure work piece	4.1 Work piece is checked and measured against specification using appropriate methods and measuring tools. 4.2 Defective work pieces are marked, recorded and reported for proper action.
5. Maintain tools, machinery and systems	5.1 Proper shutdown is carried out in accordance with standard operating procedure. 5.2 Ensure security of data, including regular back-ups and virus checks are performed as per standard operating procedure. 5.3 Basic file maintenance procedures are carried out in line with the standard operating procedure. 5.4 Systems and workplace are cleaned according to worksite procedures. 5.5 CNC milling machine is cleaned and maintained as per standard operating procedure.

	5.6 Tools, machinery and materials are cleaned and stored safely.
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Range of Variables

Variable	Range
	May include but not limited to:
1. Cutting tools and instruments	1.1 Milling Chuck 1.2 Collet 1.3 Mechanical edge finder 1.4 Z-zero setter 1.5 Pull-stud 1.6 Face/Bull Mill 1.7 End Mill 1.8 Ball mill 1.9 Tapper mill 1.10 Drill Bits 1.11 Reamers 1.12 Inserts
2. Clamping devices	2.1 Machine vice 2.2 Angle plate 2.3 V-Block, U-Clamp and C-Clamp 2.4 Step-Block 2.5 T-Slot bolt 2.6 Pneumatic fastening clamps 2.7 Jig and fixtures
3. Stages of milling	3.1 Roughing 3.2 Semi-finishing 3.3 Finishing
4. CNC milling operations	4.1 Facing 4.2 Pocket 4.3 Contour 4.4 Parallel 4.5 Scallop 4.6 Drilling 4.7 Boring

Circular content Guide

1. Underpinning knowledge	1.1 Job requirements and work piece specifications 1.2 Types of cutting tools and materials 1.3 Procedure for checking lubricant and coolant 1.4 Procedure for checking air and hydraulic pressure 1.5 Types of Clamping devices 1.6 Different stages of milling 1.7 NC-Code commands 1.8 Procedure for running CNC milling machine 1.9 Uses of measuring tools for corrective measures
2. Underpinning skills	2.1 Establishing job requirements and work piece specifications 2.2 Identifying and selecting appropriate tools and attachments 2.3 Editing NC-Code commands 2.4 Setting-up work piece to using appropriate tools and clamping device 2.5 Setting work piece to avoid distortion on release of clamping devices

	2.6 Planning and sequencing of work 2.7 Checking conformance to specifications 2.8 Performing CNC milling machine operations 2.9 Checking and measuring work piece against specifications within specified tolerances 2.10 Performing any necessary corrective measures
3. Underpinning attitudes	3.1 Prompt in carrying out activities 3.2 Tidy and punctual 3.3 Eager to learn 3.4 Active on teamwork 3.5 Sincere and honest concerning duties 3.6 Concerned for proper use of tools 3.7 Concerned about the work environment 3.8 Committed to occupational health and safety practices
4. Resource implications	The following resources must be provided: 4.1 Workplace 4.2 Computer/laptop/notebook 4.3 Software 4.4 Personal protective equipment (PPE) 4.5 Milling Tools and attachments 4.6 CNC milling machine 4.7 Drawings, specifications and work instructions 4.8 Materials

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Prepared CNC milling machine 1.2 Set-up workpiece using appropriate tools and clamping devices 1.3 Carried out three stages of milling 1.4 Performed CNC milling machine operations
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Oral test 2.3 Observation 2.4 Demonstration
3. Context of assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

Unit of Competency: CARRY OUT CNC WIRE CUT MACHINE OPERATIONS	Nominal Duration: 40 hrs.	Unit Code: SICIP-LE-CNC-04-O
Unit Descriptor: This unit of competency covers the skills, knowledge and attitudes required to Carry out CNC wire cut machine operations. It specifically includes setting- up machine, wire and work piece, loading program to wire cut machine, performing CNC wire cut operations and cleaning and maintaining wire cut machine.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set- up machine, wire and work piece	1.1 Coolant and lubricant are checked, and machine guard is adjusted 1.2 <u>Wire</u> and feed roller are set according to sequence of operations. 1.3 Clamping device is tightened as per standard operating procedure. 1.4 Work piece is mounted on clamping device using tools and instruments as per workplace guidelines.
2. Load program to wire cut machine	2.1 Engineering drawings are read and interpreted to define optimum tool path geometry. 2.2 Program is loaded into machine controller. 2.3 Program is simulated to determine the correctness of tool path and work parameters. 2.4 Program is stored as per standard operating procedure.
3. Perform CNC wire cut operations	3.1 Machine parameters such as wire offset, wire speed and power settings are set. 3.2 Machine is prepared, work piece is loaded and aligned, and data and reference points are set as per standard operating procedures. 3.3 Program is reset to ensure start position from the first program block. 3.4 Machine is operated to produce work piece/product. 3.5 Finished product is checked for conformance with specification and drawing.
4. Clean and maintain wire cut machine	4.1 Wire cut machine is cleaned. 4.2 Workplace is cleaned. 4.3 Waste materials are disposed of correctly. 4.4 Tools and finished product are stored safely pursuant to workplace guidelines.

Range of Variables

Variable	Range
	May include but not limited to:
1. Wire	1.1 Molybdenum wire

	1.2 Brass wire 1.3 Diffusion-annealed
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Circular content Guide

1. Underpinning knowledge	1.1 Types of wire 1.2 Types of Tool paths 1.3 Sequence of operations 1.4 Machine parameters 1.5 Procedure for clamping work 1.6 CNC wire cut machine operations 1.7 Maintenance procedure
2. Underpinning skills	2.1 Reading and interpreting drawings and job specifications 2.2 Calculating optimum tool path geometry 2.3 Loading and storing program 2.4 Mounting wire to the machine 2.5 Setting desired machining parameters 2.6 Setting wire reference 2.7 Mounting work piece on clamping device 2.8 Performing wire cut machine operation
3. Underpinning attitudes	3.1 Prompt in carrying out activities 3.2 Tidy and punctual 3.3 Eager to learn 3.4 Active on teamwork 3.5 Sincere and honest concerning duties 3.6 Concerned for proper use of tools 3.7 Concerned about the work environment 3.8 Committed to occupational health and safety practices
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/Laptop/Notebook 4.3 Software 4.4 Internet 4.5 Drawings, specifications and work instructions 4.6 Personal protective equipment (PPE) 4.7 Tools and equipment 4.8 CNC wire cut machine 4.9 Materials

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 Calculated optimum tool path geometry 1.2 Loaded and stored program 1.3 Mounted wire to the machine 1.4 Set wire reference 1.5 Mounted work piece on clamping device 1.6 Performed wire cut machine operation
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Oral test 2.3 Observation 2.4 Demonstration
3. Context of assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	CNC Machine Operation
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom and Workshop):

- Classroom : 350 sft
- Workshop : 800-1000 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer with relevant software	11
2.	CNC Lathe Machine	1
3.	CNC Milling Machine	1
4.	CNC Wire Cut Machine	1
5.	Tool Grinding Machine	1
6.	Radial Drill Machine	1
7.	Electric Hand Drill Machine	2
8.	Bench Vice	5
9.	Working Table	2
10.	Pedestal/Bench Grinding Machine	2

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP. For the operation of training course on CNC Machine Operation, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.